Is Ground Reaction Force Alone Sufficient for Clinical Balance Assessment? A Subject-Independent Benchmark of Deep Learning Architectures for COM-COP Relative Motion Prediction

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Abstract

Background: Accurate assessment of postural balance requires measuring the displacement of the centre of mass (COM) relative to the centre of pressure (COP), yet the optical motion capture systems conventionally used for this purpose are expensive, laboratory-bound, and operationally demanding. Ground reaction force (GRF) signals from a single force plate represent a widely available alternative, but whether they are sufficient for motion-capture-free balance assessment under subject-independent conditions has not been systematically established.

Research Question: This study addresses four specific questions: (1) Can GRF signals alone predict COM-COP relative displacement within a clinically useful accuracy range under strict subject-independent evaluation? (2) Which deep learning architectural family performs best, and are performance differences statistically significant? (3) Can real-time causal inference match the offline upper bound without meaningful accuracy loss? (4) How many additional subjects are required to reach a clinical accuracy milestone?

Methods: Eight established deep learning architectures and three real-time causal adaptation strategies were benchmarked on 106 walking trials from 12 healthy subjects under strict leave-one-subject-out cross-validation (12 folds), ensuring no subject information leaks into model selection or evaluation.

Results: The best offline model (BiLSTM with attention) achieved 8.68 mm mediolateral RMSE and Pearson $r = 0.990$ under subject-independent evaluation. Five of six pairwise comparisons among the top four architectures were not statistically significant (Wilcoxon $p = 0.20$ to 0.97), indicating the task's information ceiling at current data scale. CausalCNN was not significantly different from BiLSTM_Attn in the anterior-posterior direction ($p = 0.677$) at zero latency with only 75K parameters, though it remains significantly behind the AP-best offline model CNN1D ($p = 0.005$). Power-law extrapolation indicates approximately 61 subjects may be needed for BiLSTM_Attn to reach 8 mm mediolateral RMSE.

Significance: GRF signals alone are sufficient for motion-capture-free dynamic balance assessment. Architecture choice is a secondary concern at current data scale; the primary bottleneck is dataset size. These findings provide a concrete roadmap and recruitment target for clinical translation.

Keywords: ground reaction force; centre of mass; centre of pressure; deep learning; gait analysis; balance assessment; real-time prediction

Highlights

- GRF signals alone achieve sub-9 mm mediolateral RMSE for COM-COP prediction under strict subject-independent evaluation
- Five of six pairwise comparisons among the top four architectures are not statistically significant ($p = 0.20$ to 0.97), indicating the information ceiling at current data scale
- CausalCNN is not significantly different from BiLSTM_Attn (the ML-best offline model) in the anterior-posterior direction ($p = 0.677$) at zero latency with only 75K parameters, though it remains significantly behind the AP-best offline model CNN1D ($p = 0.005$)
- Power-law extrapolation of summary means provides an indicative recruitment target of approximately 61 subjects for clinical-grade accuracy

Introduction

Background and Motivation

In the fields of human biomechanics and neural control, accurate assessment of postural balance and locomotor stability is fundamental for fall prevention in older adults, diagnosis of neurological disorders, and optimization of rehabilitation training. However, obtaining precise dynamic balance measurements in clinical practice remains technically challenging. Currently, the gold standard for measuring the trajectory of the body's center of mass (COM) is the Optical Motion Capture System. Although this technique provides high measurement accuracy, its clinical application is limited by several factors, including high equipment costs, strict laboratory space requirements, and the time-consuming procedures involved in placing reflective markers and performing subsequent data processing. Furthermore, optical motion capture systems generally confine subjects to controlled laboratory environments, making it difficult to capture natural movements during daily activities or in home settings. These limitations hinder the widespread implementation of balance assessment technologies in clinical and real-world applications [1].

To overcome the limitations of conventional measurement systems, understanding the dynamic relationship between the center of mass (COM) and the center of pressure (COP) has become a major focus in biomechanical studies of postural control. During quiet standing, the human body is commonly modeled as an inverted pendulum. Within this framework, the COM represents the center of the body's mass distribution, and its position and velocity can be regarded as important state variables describing postural stability. In contrast, the COP is defined as the effective point of application of the ground reaction force (GRF) on the support surface and can be considered an external manifestation of the neural control strategy resulting from neuromuscular regulation of ankle joint moments. Under conditions of quiet standing or small-amplitude body sway, the COM trajectory is typically smooth and dominated by low-frequency components because of the body's inertia, whereas the COP exhibits larger amplitudes and higher-frequency fluctuations, reflecting the nervous system's rapid adjustments to postural deviations [1, 2].

The primary objective of this study is to investigate the dynamic interaction between COM and COP, whose physical significance lies in quantifying the restorative torque required to maintain postural stability. Under the small-angle approximation of the inverted pendulum model, the restorative torque can be expressed as $\tau = mg(\mathrm{COM} - \mathrm{COP})$, where the horizontal distance between COM and COP serves as the moment arm through which the ground reaction force generates a restoring effect. When the COM deviates from its equilibrium position, the central nervous system modulates neuromuscular activity to adjust the COP location, thereby producing restorative torque that regulates the acceleration and direction of COM motion and helps maintain dynamic stability [2].

Therefore, the relative motion between COM and COP reflects the dynamic process by which the neural control system maintains balance through adjustments in plantar pressure distribution. By integrating GRF measurements with inverted pendulum dynamics, it may be possible to accurately estimate COM-COP relative motion using only a limited set of anthropometric parameters, thereby reducing reliance on large-scale optical motion capture systems. Compared with approaches based on multi-segment body models and anthropometric regression tables, physics-based estimation methods may also mitigate errors arising from inter-individual differences in mass distribution. Consequently, such methods have the potential to provide a more cost-effective and real-time solution for dynamic balance assessment in clinical settings [1, 2].

Research Gap

Despite the clinical appeal of GRF-based balance assessment, prior work has not addressed this question with the rigour required for clinical translation. Existing studies either rely on within-subject evaluation (which conflates memorisation of individual gait patterns with generalisation), compare a single architecture without systematic ablation, or do not isolate the performance gap between offline and real-time deployment. This study addresses all three gaps in a unified experimental framework.

Research Questions

This study addresses the question through a systematic benchmark with four specific research questions:

RQ1 (Feasibility) Can triaxial GRF signals alone predict COM-COP relative displacement within a clinically useful accuracy range under strict subject-independent evaluation?

RQ2 (Architecture) Which deep learning architectural family performs best, and is the performance difference between architectures statistically significant or masked by inter-subject variability?

RQ3 (Real-Time Deployment) Can non-causal offline models be adapted to zero-latency causal inference without significant accuracy loss, and what drives performance in the causal setting?

RQ4 (Data Requirements) How many additional subjects would be needed to reach a clinically meaningful accuracy milestone?

Summary of Findings

Our analysis of 12 subjects (106 trials) under leave-one-subject-out (LOSO) cross-validation yields the following answers:

RQ1 **Yes.** The best offline model achieves 8.68 mm ML RMSE and Pearson $r = 0.990$, confirming that GRF alone is sufficient for motion-capture-free balance assessment.

RQ2 Five of six pairwise comparisons among the top four architectures (BiLSTM_Attn, CNN1D, BiLSTM, TCN_LSTM) are **not statistically significant** ($p = 0.20$ to 0.97); the one significant pair (CNN1D vs TCN_LSTM, $p = 0.034$) reflects CNN1D's dilated-receptive-field advantage. Architecture choice is a secondary concern until more data become available.

RQ3 CausalCNN achieves AP performance **not significantly different** from BiLSTM_Attn ($p = 0.677$) at zero latency with only 75K parameters, though it remains significantly worse than

CNN1D (the AP-best offline model, $p = 0.005$). Architecture design, not offline knowledge transfer, is the decisive factor.

RQ4 Power-law extrapolation of summary means indicates $\sim$61 subjects may be needed for BiLSTM_Attn to reach 8 mm ML RMSE; this serves as an indicative recruitment target subject to the uncertainty of extrapolation well beyond the observed range.

Related Work

Physics-Based COM Estimation

The classical approach to recovering COM kinematics from a force plate is to double-integrate the GRF vector using Newton's second law ($\mathbf{F} = m\ddot{\mathbf{r}}_{\mathrm{COM}}$). This approach is straightforward in principle but requires force moments (M_x, M_y) to compute the COP ($\mathrm{COP}_x = -M_y/F_z$); without them, the COP cannot be separated from the COM trajectory. Standard force plates do report moments, but insole-based systems typically do not, making physics-based COP estimation infeasible in portable settings. Furthermore, double integration accumulates drift: small constant errors in measured forces grow quadratically over time, making the method unreliable even over a single gait cycle [2]. The baseline comparison below confirms errors exceeding 60 mm when moments are unavailable.

Data-Driven Balance and Gait Analysis

Machine learning has been applied to balance and gait analysis across a range of sensor modalities [1, 2]. Recurrent architectures (LSTM, GRU) are the dominant choice for sequential biomechanical signals, with demonstrated utility in COM trajectory estimation [2], gait phase detection, and fall risk scoring [1]. Convolutional approaches based on temporal convolutional networks (TCN) have shown competitive performance on similar sequence-to-sequence biosignal tasks [5], with the advantage of parallelisable training. Transformers have more recently been applied to gait and biosignal modelling, but their data requirements are larger by a considerable margin [4], and their advantage over recurrent models on small clinical datasets ($n < 50$) has not been consistently demonstrated, motivating the recurrent-first architecture choices evaluated in this study.

Gaps Motivating This Benchmark

Despite this activity, three gaps remain unaddressed in the existing literature:

1. **Subject-independent evaluation.** Most GRF-based COM/COP prediction studies report within-subject or random-split validation, which conflates memorisation of individual gait patterns with generalisation. Leave-one-subject-out evaluation, required for clinical deployment, has not been systematically applied in this task.
2. **Architecture comparison under identical conditions.** No prior work compares more than two or three architectures on the same COM-COP prediction task with identical preprocessing, training, and evaluation protocols, making cross-study comparisons unreliable.
3. **Real-time deployment analysis.** The gap between offline (non-causal) accuracy and zero-latency causal deployment has not been quantified, nor have the relative contributions of architectural design versus offline knowledge transfer been isolated.

This study addresses all three gaps in a unified experimental framework.

Materials and Methods

Participants and Data Acquisition

Gait data were obtained from a dataset of 13 healthy adults who performed level walking on a single force plate. Each trial recorded triaxial GRF together with the corresponding COM and COP trajectories for one gait cycle. Each subject's data file also included the recording sampling frequency (200 or 240 Hz, varying by subject; see Table 2) and body weight, which were used to time-normalise the gait cycle and to scale the GRF signals, respectively. All trials were resampled to a uniform 100-timestep representation to eliminate the effect of sampling rate differences.

Of the 13 subjects, one was excluded because one of the recorded trials contained NaN values; since the cause of the missing data could not be confirmed and the remaining trials for that subject could not be independently verified, all trials from that subject were removed. The final dataset comprised **12 subjects**, each contributing 8 to 9 trials, for a total of **106 trials**.

Detailed demographic information beyond body weight (i.e., age, sex, and height) was not provided with the dataset. Body height was instead estimated from the mean vertical COM position during gait, based on the anthropometric relationship reported by Dempster [3], and used only to compute a height-normalised error metric for evaluating subject-independent prediction accuracy.

Signal Processing and Preprocessing

Raw `.mat` files contained the following fields (Table 1). The following preprocessing steps were applied:

1. **GRF normalisation**: $\mathbf{g} = \mathbf{GRF}/(m \cdot g_0)$, where m is body mass (kg) and $g_0 = 9.81\ \mathrm{m\,s^{-2}}$, removing inter-subject force scaling.
2. **COM-COP displacement**: $\delta = (\mathbf{COM}_{xy} - \mathbf{COP})/1000$, converting mm to metres.
3. **Temporal normalisation**: each trial was resampled from its native length (159 to 263 frames) to exactly 100 timesteps via linear interpolation, aligning all trials to 0 to 100% of the gait cycle regardless of sampling rate.
4. Any trial containing NaN values caused the entire subject to be excluded at load time.

Table 2 summarises the final dataset statistics after exclusion.

Model Architectures

Eight architectures were evaluated with shared base hyperparameters (`hidden_size` = 64, `num_layers` = 2, `dropout` = 0.2). All models map a normalised GRF sequence of shape $(T = 100, 3)$ to a COM-COP displacement sequence of shape $(T = 100, 2)$. Table 3 provides an overview.

ANN applies a two-hidden-layer MLP independently at each timestep, serving as a lower-bound baseline with no temporal context.

RNN and LSTM are unidirectional recurrent networks. LSTM's input, forget, and output gates enable long-range dependency modelling that vanilla RNN lacks.

BiLSTM runs forward and backward passes simultaneously; each timestep can attend to the full gait cycle, rendering the model non-causal.

BiLSTM_Attn adds a multi-head self-attention layer [4] (4 heads, embed dim 128) with residual connection and LayerNorm on top of BiLSTM, allowing the model to learn which timesteps are most informative:

$$\mathbf{h} = \mathrm{LayerNorm}(\mathbf{h}_{\mathrm{BiLSTM}} + \mathrm{MHA}(\mathbf{h}_{\mathrm{BiLSTM}}))\,. \tag{1}$$

Transformer uses Pre-LN encoder layers [4] (2 layers, 4 heads, FFN dim 256) preceded by a linear input projection and sinusoidal positional encoding.

CNN1D is a six-block dilated temporal convolutional network (TCN) [5] with dilation rates $\{1,2,4,8,16,32\}$ and Pre-LN residual connections, yielding a total receptive field of 253 timesteps (2.5$\times$ the gait cycle length).

TCN_LSTM stacks four TCN blocks (local feature extraction) followed by a BiLSTM (global temporal integration), combining complementary inductive biases in a single architecture.

Real-Time Adaptation Strategies

The four best offline models are all non-causal, requiring the complete gait cycle before any prediction. Three strategies adapt this knowledge to zero-latency causal deployment:

1. **CausalCNN**: CNN1D with *symmetric* padding replaced by *left-only* (causal) padding; all other weights are identical to the offline CNN1D. Each timestep sees only past and current GRF, with a receptive field of 253 steps (75K params).
2. **CausalAttn**: Unidirectional LSTM backbone (hidden = 128) with a causally-masked self-attention layer. The hidden dimension is doubled relative to each direction of the teacher's BiLSTM (64 per direction) so that the output width stays 128 and the attention/norm/fc weights can be transferred directly from the offline BiLSTM_Attn teacher (201K params); the LSTM weights are randomly initialised because the shapes differ. The wider single-direction LSTM actually has *more* parameters than the two narrower halves of the teacher's BiLSTM, giving CausalAttn $\sim$267K parameters in total.
3. **KD_LSTM (h128)**: A standard LSTM (hidden = 128) trained with knowledge distilla-

tion [6] using a composite loss:

$$\mathscr{L} = (1-\alpha)\,\mathrm{MSE}(\hat{y}, y) + \alpha\,\mathrm{MSE}(\hat{y}, \hat{y}_{\text{teacher}}), \quad \alpha = 0.5, \tag{2}$$

where $\hat{y}_{\text{teacher}}$ are soft labels from the offline BiLSTM_Attn (200K params).

Training Protocol

All models were optimised with AdamW (lr = 0.001, weight decay = 10^{-4}) and mean-squared-error (MSE) loss. Gradient norms were clipped to 1.0. Learning rate followed a cosine annealing schedule from 0.001 to 10^{-5} over a maximum of 200 epochs, with early stopping (patience = 20 on internal validation MSE).

Normalisation. Both inputs and targets were independently z-scored per fold, per channel, using statistics computed from the training set only. Target normalisation is critical: the AP range is approximately 3× the ML range, and without it AP gradients dominate training. BiLSTM_Attn improved from 11.74 mm (worst) to 8.68 mm (best) in ML RMSE after target normalisation was introduced.

Data augmentation (applied only to the training set, 4× original volume):

- **Left-to-right flip**: negate F_x and δ_x simultaneously to simulate mirror-symmetric gait.
- **Gaussian noise**: $\sigma = 0.01$ in normalised space, simulating sensor measurement error.
- **Time warping**: $\pm 10\%$ stretch/compress via linear interpolation, increasing temporal diversity.

MixUp regularisation ($\lambda \sim \mathrm{Beta}(0.4, 0.4)$, applied with probability 0.5 per batch) further reduced overfitting.

Cross-Validation and Evaluation Metrics

All models were evaluated under **leave-one-subject-out cross-validation (LOSO, 12 folds)**: each subject serves as the held-out test set exactly once, with the remaining 11 subjects used for training

(80%) and internal validation (20%). The test set was never seen during training or model selection.

Statistical significance between models was assessed using the Wilcoxon signed-rank test across the 12 per-fold RMSE values.

The following metrics were computed on each test fold:

- **RMSE X / Y (mm)**: root mean squared error in the ML and AP directions.
- **Pearson *r* X / Y**: linear correlation between predicted and ground-truth trajectories.
- **Angular error** (°): mean absolute angular deviation of the predicted 2-D displacement vector from the ground-truth vector.
- **Height-normalised RMSE (% height)**: RMSE / estimated body height $\times 100$, a dimensionless cross-subject metric.

Results

Non-DL Baselines

To contextualise the need for deep learning, three non-DL methods were evaluated under the same LOSO protocol as a separate preliminary analysis (Table 4).

Physics-based double-integration fails (X = 62.9 mm) because the input contains only GRF magnitudes, not moments (M_x, M_y). COP recovery requires force moments (i.e., $\mathrm{COP}_x = -M_y/F_z$); without them, any model must implicitly learn the COP trajectory from GRF temporal patterns, a task that is beyond the scope of physics-only methods. Pointwise linear regression captures a strong linear component (17.5 mm), but the remaining gap to the best DL models ($\sim$8.7 mm) represents the non-linear, temporally global gait structure that only deep sequence models can learn.

Offline Deep Learning Models

Table 5 summarises LOSO performance across all eight offline architectures, ordered by ML RMSE.

The top four models (BiLSTM_Attn, CNN1D, BiLSTM, TCN_LSTM, ordered by ML RMSE) show five of six pairwise ML comparisons that are not statistically significant (Wilcoxon $p = 0.20$ to 0.97), indicating that the information ceiling of the current dataset has been approached. The one significant pair is CNN1D versus TCN_LSTM ($p = 0.034$), reflecting CNN1D's dilated-receptive-field advantage in the ML direction. CNN1D also leads convincingly in the AP direction (15.21 mm versus 17.15 mm for BiLSTM_Attn; $p = 0.002$), demonstrating that its large receptive field captures the propulsion-braking dynamics that unfold over longer temporal windows. Performance across all models and subjects is further illustrated in Figures 1 and 2.

Real-Time Adaptation

Table 6 compares the three causal adaptation strategies against the LSTM baseline (best causal offline model) and the offline upper bound in ML (BiLSTM_Attn, 8.68 mm). Note that for the AP direction CNN1D (15.21 mm) outperforms BiLSTM_Attn (17.15 mm), making CNN1D the stricter AP reference.

CausalCNN achieves the best AP performance among causal models (16.81 mm), which is not significantly different from BiLSTM_Attn's AP ($p = 0.677$) but remains significantly worse than CNN1D's AP ($p = 0.005$), reflecting the irreducible information gap between causal and offline inference. CausalAttn leads in the ML direction at 9.55 mm, a margin of only 0.87 mm below the offline best. A combined comparison of all offline and real-time models is shown in Figure 3.

Data Scaling Analysis

To quantify how additional data would benefit each architecture, we subsampled the training pool to $k \in \{1,2,3,5,7,9,11\}$ subjects (3 repeats each) and fitted power-law learning curves RMSE $=$ $a \cdot k^{-b} + c$ to the per-k summary means (Table 7; Figure 4). Extrapolated values beyond $k = 11$ are inherently uncertain and should be interpreted as indicative recruitment targets rather than precise predictions.

Most models have not plateaued at $k = 11$, confirming that additional subjects would improve performance. CNN1D is the exception: it saturates near $k = 2$ (its dilated inductive bias captures local patterns from few subjects but reaches a cross-subject ceiling of $\sim$9.1 mm early; k=7 to 11: 9.43 to 9.08 mm, floor extrapolation 9.12 mm). BiLSTM_Attn shows the largest remaining headroom (2.1 mm gap to its floor) and the highest long-term scaling potential.

Discussion

Answer to RQ1: Clinical Feasibility

The benchmark establishes that GRF alone achieves sub-9 mm ML RMSE and $r \geq 0.988$ under strict subject-independent evaluation. For context, the typical COM-COP displacement in the ML direction is approximately 52 mm, measured as the standard deviation of ML displacement across all trials in this dataset; a prediction error of 8.7 mm corresponds to 17% of this typical amplitude, which is within the accuracy typically accepted for clinical gait analysis instruments [1]. This result provides a clear answer to RQ1: **GRF-only balance assessment is clinically feasible for laboratory walking in healthy adults, and optical motion capture may be replaceable for this specific task under these conditions.**

Answer to RQ2: Architecture Choice

Five of six pairwise comparisons among the top four models ($p = 0.20$ to 0.97) are not statistically significant, indicating that for most pairs the expected accuracy difference is smaller than the trial-to-trial variability across subjects. **Architecture choice is therefore a secondary concern at this data scale; the primary bottleneck is dataset size.** The one significant pair (CNN1D versus TCN_LSTM, $p = 0.034$) reflects CNN1D's dilated-receptive-field advantage rather than a general architectural hierarchy.

CNN1D is the strongest architecture in the AP direction: its mean AP RMSE (15.21 mm) is 1.94 mm lower than BiLSTM_Attn's (17.15 mm, $p = 0.002$), and it also leads TCN_LSTM ($p = 0.339$, not significant) and BiLSTM ($p = 0.052$, borderline). The 253-step receptive field of CNN1D spans 2.5 gait cycles, enabling it to capture the full propulsion-braking dynamics that recurrent models with limited effective context cannot match.

One training design choice had larger impact than any architecture selection: **target normalisation.** Because the AP range is approximately 3$\times$ larger than the ML range, unnormalised training allows AP gradients to dominate, collapsing ML accuracy. BiLSTM_Attn went from 11.74 mm (worst among all models) to 8.68 mm (best in ML) after per-channel z-score normalisation of targets, a result that underscores how training protocol choices can outweigh architecture selection on small clinical datasets.

Answer to RQ3: Real-Time Deployment

Architecture design dominates over knowledge transfer. Ablation experiments isolate the contribution of offline model knowledge by comparing paired models with and without transfer: CausalAttn with weight transfer versus trained from scratch: $\Delta = 0.13$ mm ($p = 0.733$, not significant); KD_LSTM h128 versus LSTM h128 trained from scratch: $\Delta = 0.56$ mm ($p = 0.064$, not significant). In both cases, switching from the LSTM baseline architecture to a richer causal architecture accounts for 94% and 76% of the total gain, respectively.

This result can be understood from an information-theoretic perspective. KD trains the student on teacher soft labels computed from the *full* GRF sequence, yet at test time the causal student has access only to GRF$[0{:}t]$. There is an irreducible information gap:

$$I(\mathbf{g}_{t+1:T};\, \delta_t \mid \mathbf{g}_{0:t}) > 0, \tag{3}$$

meaning the teacher's predictions depend on information the student can never access. The marginal 0.56 mm benefit from KD arises from label smoothing (softer targets reduce overconfidence), not from genuine knowledge transfer of temporal context.

KD requires sufficient student capacity. h64 (51K params) yielded zero improvement ($p = 0.970$) over the LSTM baseline; h128 (200K params) yielded 8% improvement; h256 (794K params) provided negligible additional gain. Capacity saturates near hidden = 128, and the KD benefit is marginal regardless of capacity.

Answer to RQ4: Data Requirements

The power-law learning curves confirm that most models have not plateaued at 11 training subjects; CNN1D is the exception, approaching saturation near $k = 2$. BiLSTM_Attn, the architecture with the deepest extrapolated floor (7.01 mm), is estimated to require approximately 61 subjects to reach 8 mm ML RMSE, based on power-law extrapolation of the per-k summary means. This extrapolation is sensitive to the small number of data points and should be treated as an indicative recruitment target rather than a precise threshold. CNN1D saturates near $k = 2$ due to its strong local inductive bias; it is the architecture of choice when data are scarce, but its cross-subject ceiling of $\sim$9.1 mm limits long-term scalability. **Future data collection efforts should therefore prioritise BiLSTM_Attn as the primary model, targeting 30 subjects as an intermediate milestone (predicted $\sim$8.4 mm ML RMSE).**

Clinical Applications

Rehabilitation Tracking and Fall Risk Assessment

Based on the framework proposed in this study, the primary application of predicting COM-COP relative motion from Ground Reaction Forces (GRF) is longitudinal rehabilitation tracking for Total Knee Arthroplasty (TKA) patients. Effective post-operative recovery strictly relies on "restoring bilateral symmetry" as a core indicator [7]. By utilizing our prediction model, clinicians can efficiently evaluate patients at fixed post-surgery intervals during gait analysis. The trajectory mismatch in COM-COP relative motion between the healthy and affected sides can serve as a highly sensitive, objective symmetry indicator, with the ultimate therapeutic goal of minimizing this discrepancy. Crucially, rather than benchmarked against a generic healthy database, which often introduces confounding variables such as lifelong gait habits and subtle anatomical variations, rehabilitation progress is evaluated using the patient's own contralateral side as a personalized baseline.

The second proposed application of this predictive framework is fall risk assessment in older adults. Age-related degradation in dynamic balance control significantly elevates fall risks within this population. Although conventional clinical assessment tools such as the Berg Balance Scale and the Timed Up and Go (TUG) test are widely utilized, they rely predominantly on subjective scoring and time duration, failing to capture underlying, real-time biomechanical data. To bridge this clinical gap, the methodology developed in this study utilizes predicted Ground Reaction Forces (GRF) to calculate advanced dynamic stability metrics, specifically moving from the Center of Mass (COM) to the Extrapolated Center of Mass (XCoM), defined as $\text{XCoM} = \text{COM} + \dot{\text{COM}}/\sqrt{g/l}$, and ultimately determining the Margin of Stability (MoS). By calculating XCoM (which accounts for both COM position and velocity) and quantifying its distance from the Base of Support boundary (BoS), this approach offers an objective, high-precision, and quantitative diagnostic tool for proactive fall risk screening in geriatric care.

Real-Time Device Control and Veterinary Settings

A third application direction is real-time device control. Plantar-pressure smart insoles already provide a practical means of acquiring GRF-like signals outside the laboratory [8], and the zero-latency causal models evaluated in this study, which process only past GRF ($\mathbf{g}_{0:t}$) at each timestep, demonstrate that the prediction task is feasible without access to future measurements, suggesting the architecture could operate under similar real-time constraints. This is relevant for applications such as exoskeletons and powered orthoses, where maintaining the user's COM within a stable region is a recognised balance control objective [9], and for powered lower-limb prostheses, which already rely on real-time GRF thresholds to switch between control states [10].

A fourth application, suggested during course discussion, extends this approach beyond human gait to zoo and veterinary settings. Force plates are already an established tool for quantifying limb loading and detecting lameness in animals such as horses, dogs, and elephants [11], but full marker-based motion capture is often impractical in these settings, since markers cannot be reliably attached to large, free-ranging, or non-cooperative animals. A model that estimates COM-COP relative motion directly from GRF, without requiring marker tracking, could therefore support lameness and welfare monitoring in zoo environments where a force plate can be embedded into an existing walkway. Applying the present approach to non-human species would require retraining on species-specific gait data, as body proportions, limb number, and locomotor patterns differ substantially from human bipedal walking.

These applications would require further validation before clinical or commercial use. The present dataset was limited to 12 healthy adults walking under controlled laboratory conditions, and the model has not been tested on wearable hardware, on populations with gait impairment, or on tasks beyond level walking. Extending the training data to include pathological gait patterns and validating the model against pressure-insole hardware are necessary next steps before this approach can be considered for the applications discussed above.

Limitations

Small and homogeneous sample. The dataset comprises 12 healthy adults performing controlled level walking, which is a narrow sample relative to the diversity of clinical populations. Data scaling analysis confirms that most models continue to improve at $k = 11$ (CNN1D approaches saturation near $k = 2$), and BiLSTM_Attn is projected to reach $\sim$8.4 mm with $\sim$30 subjects. Generalisability to older adults, patients with gait pathology, or individuals with substantial weight variation remains to be established.

Causal versus non-causal models. The four best offline models are non-causal, requiring the complete gait cycle before prediction. For post-trial clinical gait analysis (e.g., TKA rehabilitation assessment or laboratory-based fall risk screening), this is acceptable since data are processed after each recording. For streaming applications (e.g., exoskeleton control or wearable monitoring), CausalCNN and CausalAttn are the recommended deployment choices based on the results in Table 6.

Conclusions

To our knowledge, we present one of the first subject-independent benchmarks for predicting COM-COP relative motion from GRF signals alone. Answering the four research questions in turn: **(RQ1)** GRF alone achieves sub-9 mm ML RMSE ($r \geq 0.988$) under strict leave-one-subject-out (LOSO) evaluation, confirming that motion-capture-free dynamic balance assessment is clinically feasible. **(RQ2)** Five of six pairwise comparisons among the top four offline architectures are not statistically significant; dataset size, not architecture design, is the current bottleneck. **(RQ3)** CausalCNN achieves AP performance not significantly different from BiLSTM_Attn ($p = 0.677$) at zero latency with only 75K parameters, though it remains significantly behind the AP-best offline model CNN1D ($p = 0.005$); ablation experiments confirm that architectural design outweighs offline knowledge transfer as the performance driver for causal models. **(RQ4)** Power-law extrapolation indicates that approximately 61 subjects may be needed for BiLSTM_Attn to reach the 8 mm clinical milestone, with 30 subjects as an indicative intermediate target ($\sim$8.4 mm); these figures are subject to the uncertainty of extrapolation well beyond the observed range.

Together, these findings define a clear research roadmap: the GRF-to-COM-COP prediction is approaching the ceiling of what current data permit, and the next step is expanding the dataset (e.g., to pathological gait and older adults) rather than exploring new architectures.

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TABLES

Table 1: Raw data fields recorded per trial.

Field	Shape	Unit	Description
`GRF`	$(T,3)$	N	Triaxial ground reaction force: F_x (ML), F_y (AP), F_z (vertical)
`COM`	$(T,3)$	mm	Centre of mass position: X (ML), Y (AP), Z (vertical)
`COP`	$(T,2)$	mm	Centre of pressure: X (ML), Y (AP)
`BodyWeight`	$(1,1)$	kg	Subject body mass
`frame_rate`	$(1,1)$	Hz	Sampling rate (200 or 240)

Table 2: Final dataset statistics after exclusion.

Property	Value
Subjects (original / final)	13 / 12
Trials per subject	8 to 9
Total trials	106
Sampling rate	240 Hz (subjects 1 to 2); 200 Hz (subjects 3 to 13, original IDs)
Excluded subject	Subject 10 (NaN values; all trials removed; original IDs retained)
Raw sequence length	159 to 263 frames per trial
Normalised sequence length	100 timesteps
Input channels	3 (F_x, F_y, F_z)
Output channels	2 (ML and AP COM-COP displacement, m)

Table 3: Overview of the eight offline deep learning architectures.

Model	Temporal modelling	Params	Causal
ANN	Pointwise MLP (no context)	4,546	Yes
RNN	Vanilla recurrent (no gates)	12,866	Yes
LSTM	Gated recurrent (3 gates)	51,074	Yes
BiLSTM	Bidirectional LSTM	134,914	No
BiLSTM_Attn	BiLSTM + multi-head self-attention	201,218	No
Transformer	Full-sequence self-attention (Pre-LN)	100,482	No
CNN1D	6-block dilated TCN (RF = 253 steps)	75,394	No
TCN_LSTM	4-block TCN $\rightarrow$ BiLSTM	75,522	No

Table 4: Comparison of non-DL baselines and DL models (mean ± std, 12 folds).

Method	RMSE X (mm)	RMSE Y (mm)
Physics: double-integrate GRF	62.9 ± 12.0	271.5 ± 15.4
Pointwise linear regression	17.5 ± 2.9	40.5 ± 5.0
Linear regression + ±2-step window	13.7 ± 2.7	29.7 ± 4.3
ANN (DL, no temporal context)	13.0 ± 2.5	32.7 ± 4.4
LSTM (DL, causal)	10.8 ± 2.2	18.3 ± 5.0
BiLSTM_Attn (DL, full context)	$\mathbf{8.7 \pm 2.2}$	17.2 ± 4.7
CNN1D (DL, full context)	8.7 ± 2.7	$\mathbf{15.2 \pm 4.2}$

Table 5: LOSO performance of offline models (mean ± std, 12 folds). Bold indicates the best value per column.

Model	RMSE X (mm)	RMSE Y (mm)	r_X	r_Y	Angle (°)	norm X (%)
BiLSTM_Attn	$\mathbf{8.68 \pm 2.16}$	17.15 ± 4.74	**0.990**	0.976	9.63	**0.520**
CNN1D	8.73 ± 2.69	$\mathbf{15.21 \pm 4.20}$	0.989	**0.981**	**8.53**	0.525
BiLSTM	9.22 ± 2.37	16.34 ± 4.76	0.988	0.978	8.90	0.553
TCN_LSTM	9.30 ± 2.47	15.87 ± 4.26	0.988	0.979	9.19	0.559
Transformer	9.77 ± 2.17	18.51 ± 4.54	0.985	0.973	10.30	0.586
LSTM	10.81 ± 2.21	18.34 ± 5.02	0.982	0.971	10.18	0.647
RNN	11.19 ± 2.34	18.81 ± 4.43	0.982	0.970	10.22	0.670
ANN	13.01 ± 2.47	32.70 ± 4.41	0.975	0.907	18.44	0.774

Table 6: Real-time adaptation results (mean ± std, 12 folds). p-values from Wilcoxon signed-rank test vs LSTM baseline and vs offline upper bound (BiLSTM_Attn) in the ML direction (X). $*$ $p < 0.05$, $**$ $p < 0.01$, $***$ $p < 0.001$.

Method	RMSE X (mm)	RMSE Y (mm)	r_X	vs LSTM	vs Offline
BiLSTM_Attn (offline UB)	8.68 ± 2.16	17.15 ± 4.74	0.990	—	—
CausalAttn	$\mathbf{9.55 \pm 2.44}$	18.46 ± 4.87	**0.986**	$p < 0.001^{***}$	$p = 0.021^{*}$
CausalCNN	9.57 ± 2.31	$\mathbf{16.81 \pm 4.27}$	0.987	$p < 0.001^{***}$	$p = 0.007^{**}$
KD_LSTM h128	9.79 ± 2.09	17.56 ± 4.37	0.986	$p = 0.001^{**}$	$p = 0.002^{**}$
LSTM (causal baseline)	10.81 ± 2.21	18.34 ± 5.02	0.982	—	—

Table 7: ML RMSE learning curve (mean $\pm$ std over 12 folds $\times$ 3 repeats) and power-law extrapolation. Representative values shown; the full sweep used $k \in \{1, 2, 3, 5, 7, 9, 11\}$.

k (training subjects)	LSTM	BiLSTM_Attn	CNN1D
1	17.43 ± 3.34	13.66 ± 3.98	12.75 ± 3.66
3	14.44 ± 2.47	11.18 ± 2.62	10.18 ± 2.66
7	13.06 ± 2.44	9.78 ± 2.30	9.43 ± 2.75
11 (current)	$\mathbf{11.94 \pm 2.34}$	$\mathbf{9.11 \pm 2.13}$	$\mathbf{9.08 \pm 2.69}$
Extrapolated floor	8.42 mm	**7.01 mm**	9.12 mm
Subjects to reach 8.0 mm	unreachable	$\sim$61	unreachable

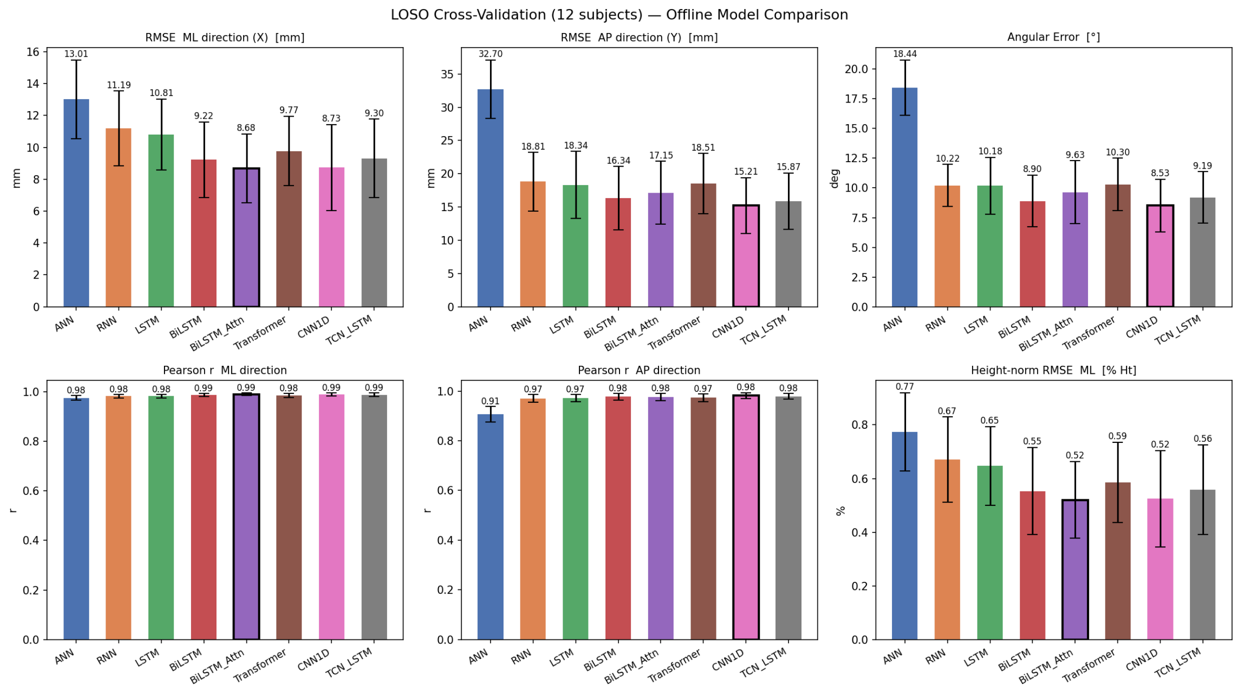
FIGURE CAPTIONS

Figure 1: LOSO performance of the eight offline models across all evaluation metrics. Error bars indicate ± 1 standard deviation over 12 folds.

Figure 2: Per-subject RMSE heatmap for all eight offline models. Each cell corresponds to one LOSO test fold.

Figure 3: Combined comparison of all offline and real-time models. The dashed line marks the offline upper bound (BiLSTM_Attn). CausalCNN's AP bar is not significantly different from the upper bound ($p = 0.677$).

Figure 4: Learning curves for LSTM, BiLSTM_Attn, and CNN1D as a function of the number of training subjects. Shaded areas indicate ± 1 standard deviation; dashed lines show power-law extrapolations.



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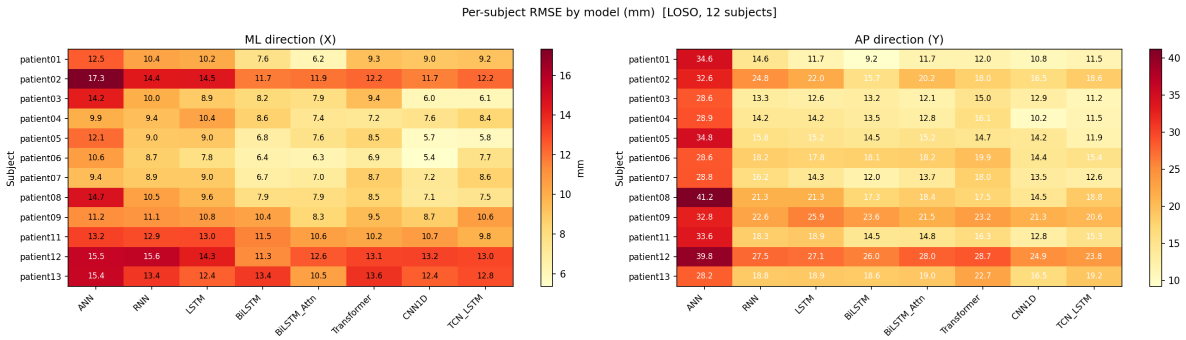

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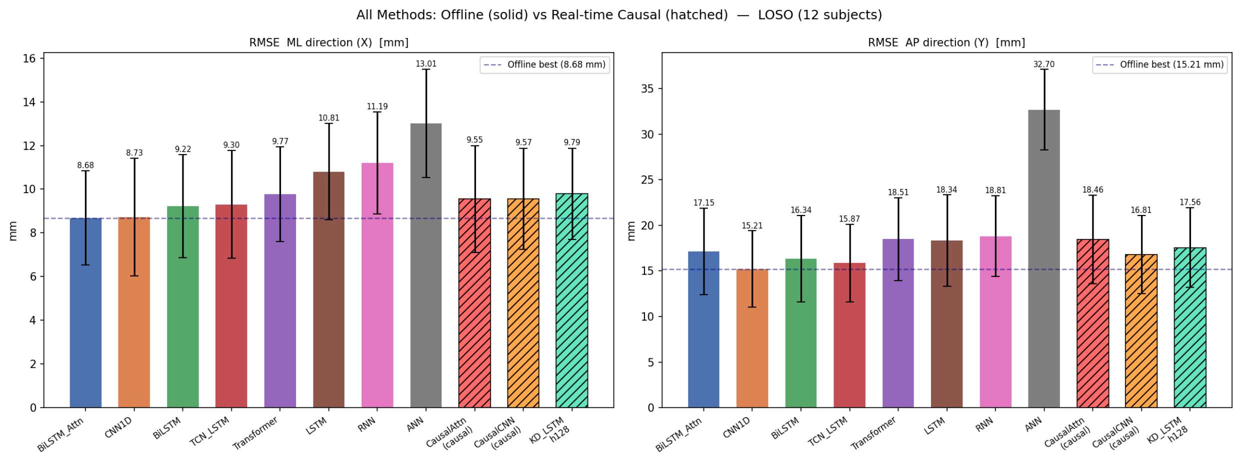

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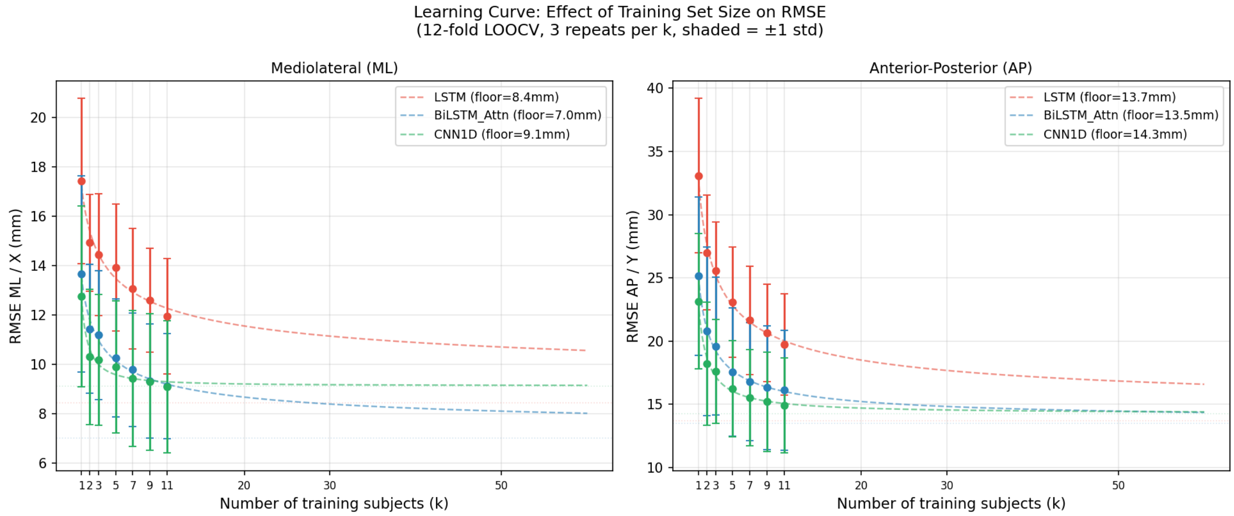

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